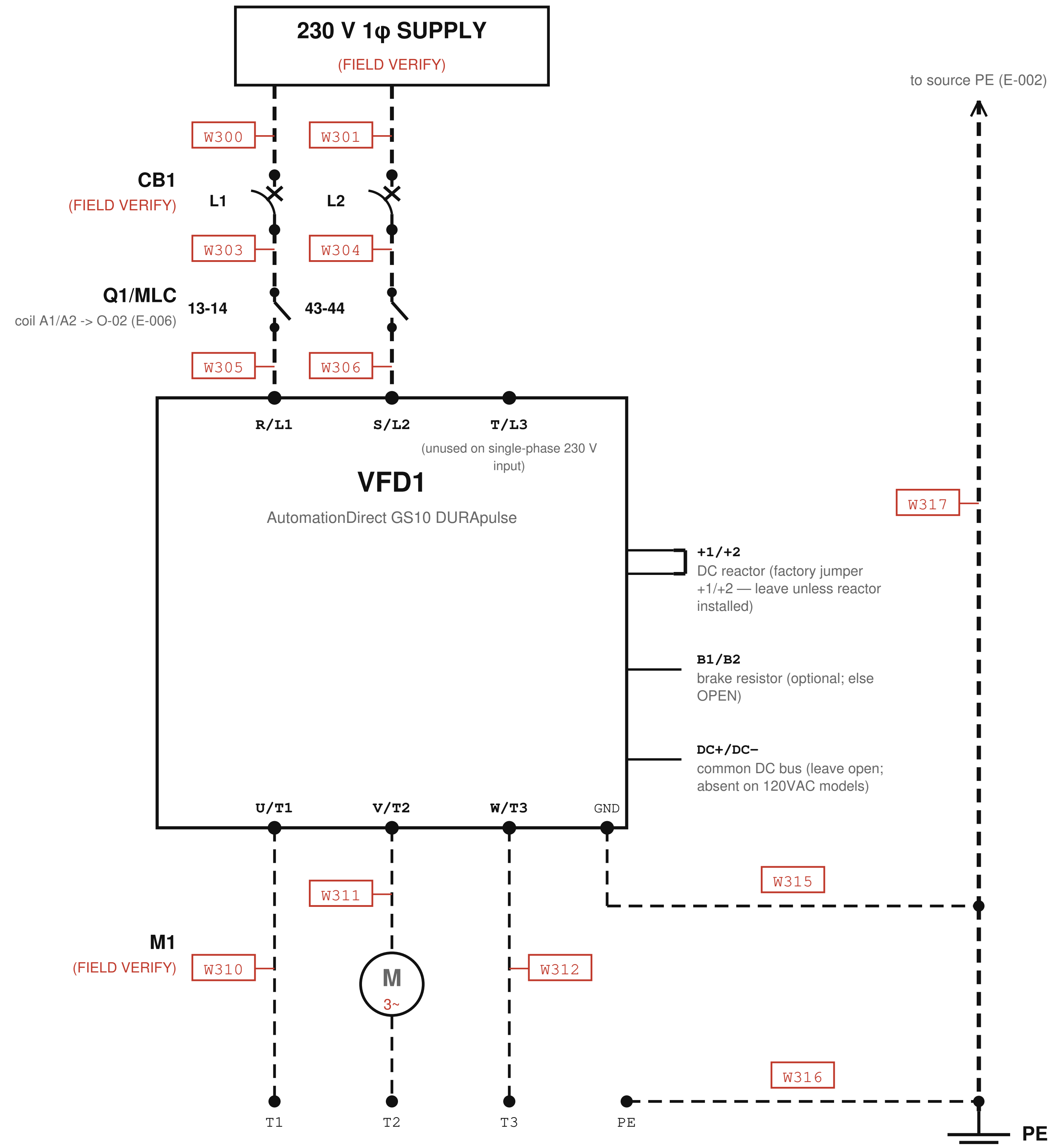
E-003 VFD POWER

Conv_Simple / CV-101 (garage conveyor) — supply → CB1 → Q1/MLC (supply switch) → VFD1 (GS10) → M1 · 230 V 1φ in → 230 V 3φ to motor · every conductor FIELD VERIFY



CONNECTION TABLE

Wire	From	To	Signal	Type	Status	Notes
W300	SUPPLY (E-002)	CB1.1	L1 230V 1ph supply	power_line	field_verify	
W301	SUPPLY (E-002)	CB1.3	L2 230V 1ph supply	power_line	field_verify	
W303	CB1.2	Q1.13	L1 protected -> MLC	power_line	field_verify	
W304	CB1.4	Q1.43	L2 protected -> MLC	power_line	field_verify	
W305	Q1.14	VFD1.R/L1	L1 switched -> drive	power_line	field_verify	
W306	Q1.44	VFD1.S/L2	L2 switched -> drive	power_line	field_verify	
W310	VFD1.U/T1	M1.T1	motor phase U	motor_lead	field_verify	
W311	VFD1.V/T2	M1.T2	motor phase V	motor_lead	field_verify	
W312	VFD1.W/T3	M1.T3	motor phase W	motor_lead	field_verify	
W315	VFD1.GND	PE bus	drive PE (≤0.1Ω)	ground	field_verify	
W316	M1.PE	PE bus	motor frame PE	ground	field_verify	
W317	PE bus	SUPPLY (E-002)	PE to source	ground	field_verify	

Supply is CONFIRMED 230 V single-phase (technician-confirmed 2026-07-11, review/PHOTO_EVIDENCE_V7.md) — R/L1, S/L2 only; T/L3 unused. Only the GS10 exact model/frame, breaker rating, and wire gauge remain NOT DOCUMENTED — every conductor FIELD VERIFY.
P00.01 = 1.60 A (2026-05-20 export) is the only sizing clue.
MLC (Q1) is a control relay USED AS the single-phase drive-supply switch — its 2 NO contacts (13-14, 43-44) carry the 230 V 1φ supply into the GS10; coil A1/A2 from O-02 (E-006). Energizing O-02 powers the drive (technician-confirmed 2026-07-11).

SAFETY

- Never start/stop via input power (L1811-1813); CB1 is branch protection only, not a run/stop device.
- LOTO + wait ≥5 min for DC-bus discharge before touching (WI p.2 §2).
- Monitored e-stop is NOT a safety stop — hardwire S0 to remove drive power per NFPA 79 / EN 60204-1.
- PE resistance ≤0.1Ω (L1760-61); shield/conduit bonded both ends (L1792-95); RFI jumper in/out FIELD VERIFY (L1693-1718, OI-17).

LEGEND

- VERIFIED (solid)
- FIELD VERIFY (dashed)
- Wxxx proposed wire number (red = unverified)

NOTES

- CB1 REQUIRED per GS10_UM L1758-1759 — type/rating unknown (OI-15).
- M1: swap any two motor leads to reverse — GS10_UM L1773-1776.
- VFD control source = Modbus — see E-007. No control wiring on this sheet.
- Supply is 230 V SINGLE-PHASE (2-wire); drive output = 230 V 3φ to a 230 V motor (technician-confirmed 2026-07-11).
- GS10 manual also warns against cycling a POWER-CIRCUIT switching device for normal run/stop — emergency situations only (GS10_UM.txt L1754-1757), same caution as L1811-1813 above. This applies to Q1/MLC (the device that powers the drive) exactly as it applies to CB1: normal run/stop is via Modbus (E-007), never by cycling Q1's coil.

SOURCES:

plc/GS10_UM.txt (terminals L1971-1986; topology L1750-1813) ·
plc/GS10_Integration_Guide.md (Q1 phase-5 test)
plc/Prog_init_ConvSimple_v2.1.st (vfd_run_permit) · plc/MIRA_PLC_WorkInstruction_v3.pdf (LOTO §2)
photos review/photos/wire_1..4.jpg (2026-07-11, actual devices)
review/PHOTO_EVIDENCE_V6.md (technician field correction 2026-07-11 — single-phase supply topology, MLC in power path)
review/PHOTO_EVIDENCE_V7.md (technician confirmation 2026-07-11 — 230 V motor output, OI-27 resolved)

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Drawn: MIRA / FactoryLM terminals per GS10 UM 1st Ed Rev B	REV A 2026-07-11	3 of 9